

WINNER ENGINEERING LEARNING HUB SERIES

FUNDAMENTAL TOPIC

#03 Pipe Connection



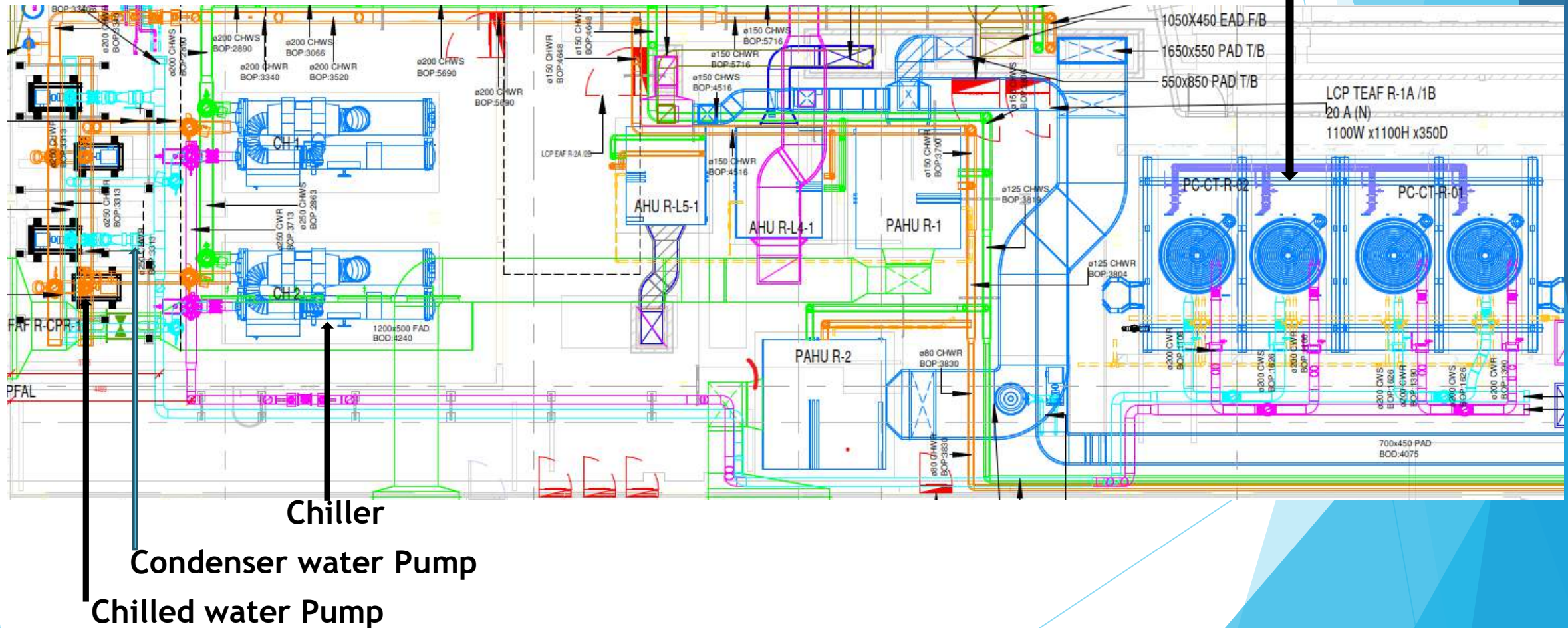
Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service

Table of Content

- ▶ Chiller Plant Room
 1. Chiller
 2. Auto Tube Cleaning System for Chiller
 3. Cooling Tower
 4. Chemical Water Treatment for Cooling Tower
 5. Pump
- ▶ AHU & HRU Pipe connection

Chiller Plant Room

- ▶ Chiller plant room is a dedicated space within a building where the equipment and systems for cooling and air conditioning are housed.
- ▶ It is typically located in the basement or Rooftop.



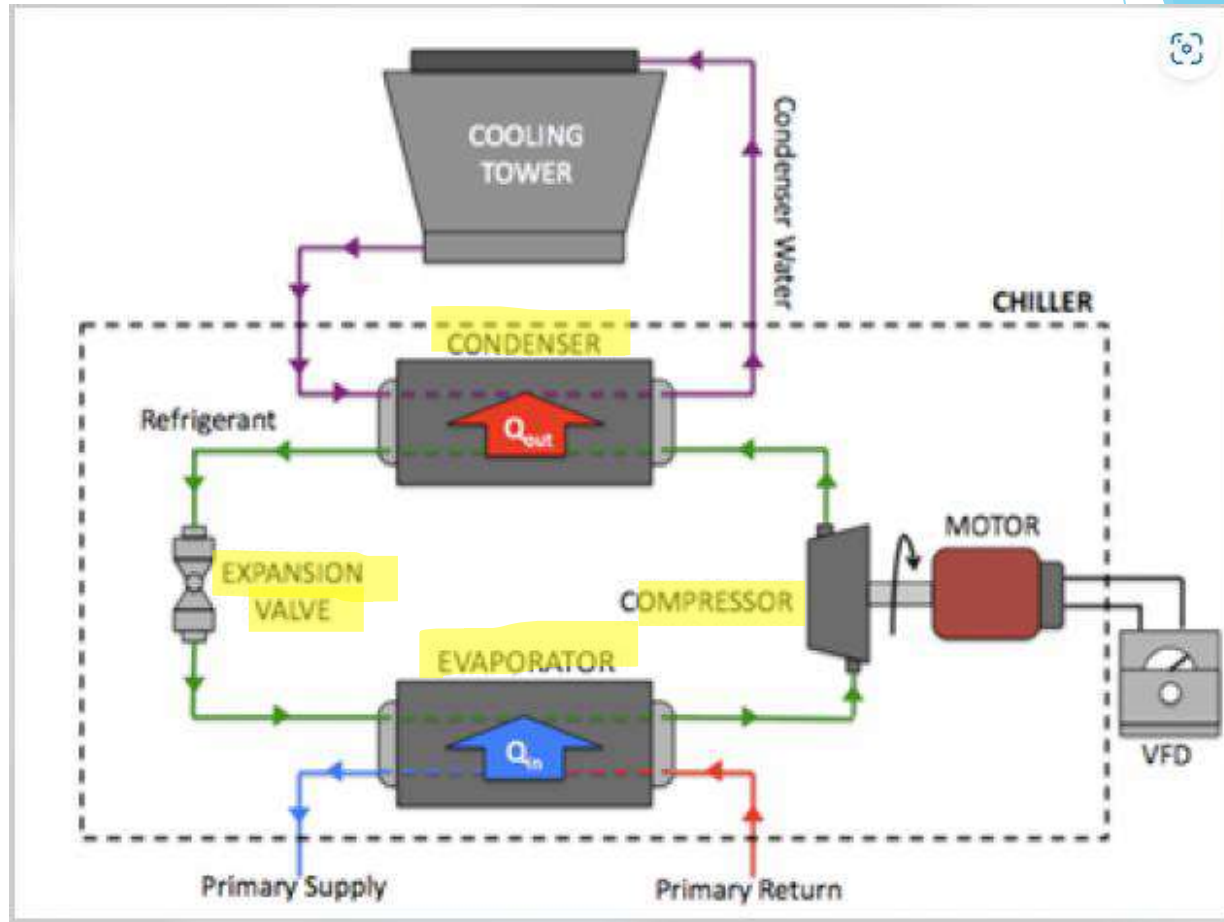
1. Chiller

There are 4 main parts in the Chiller:

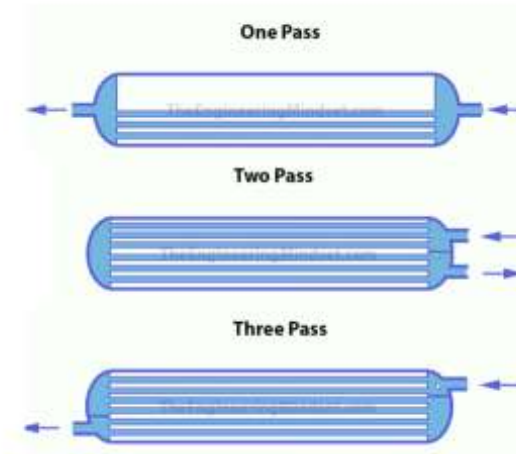
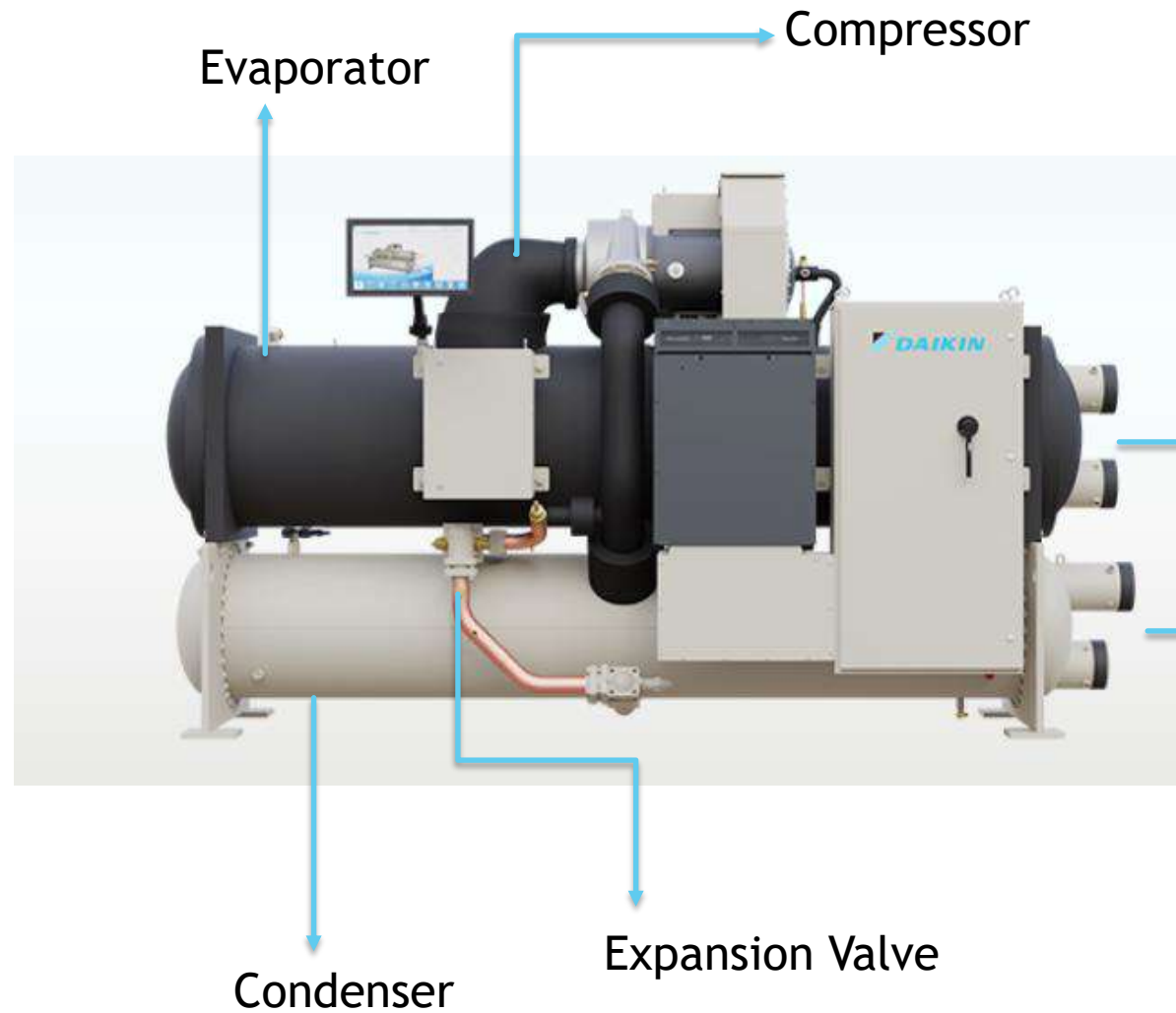
1. Evaporator
2. Condenser
3. Expansion Valve
4. Compressor

And connected with 3 different Loop:

1. Chilled water loop
2. Condenser Water Loop
3. Refrigerant Loop



Main Component of the Chiller



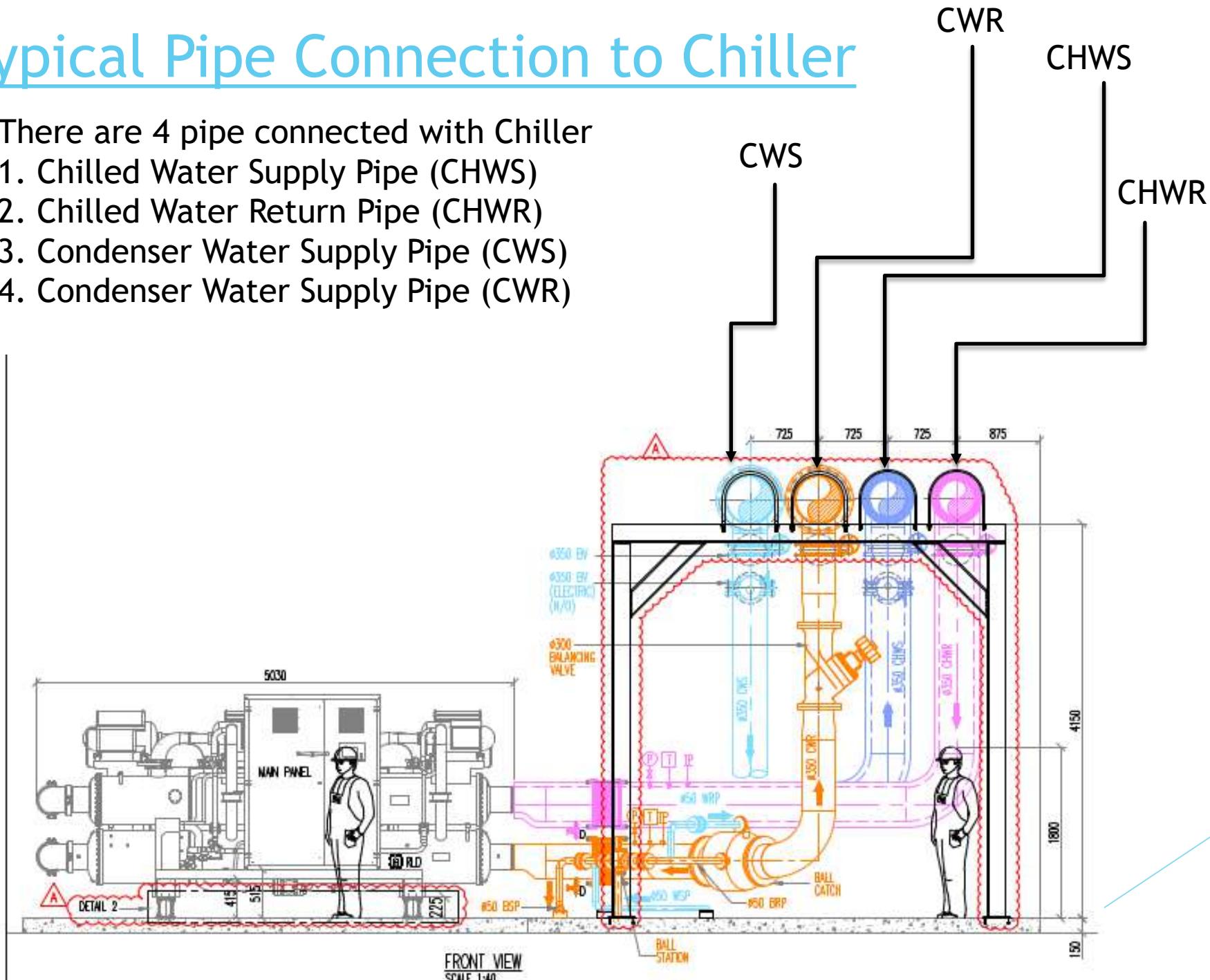
Chilled water
inlet & outlet

Condenser Water
inlet & outlet

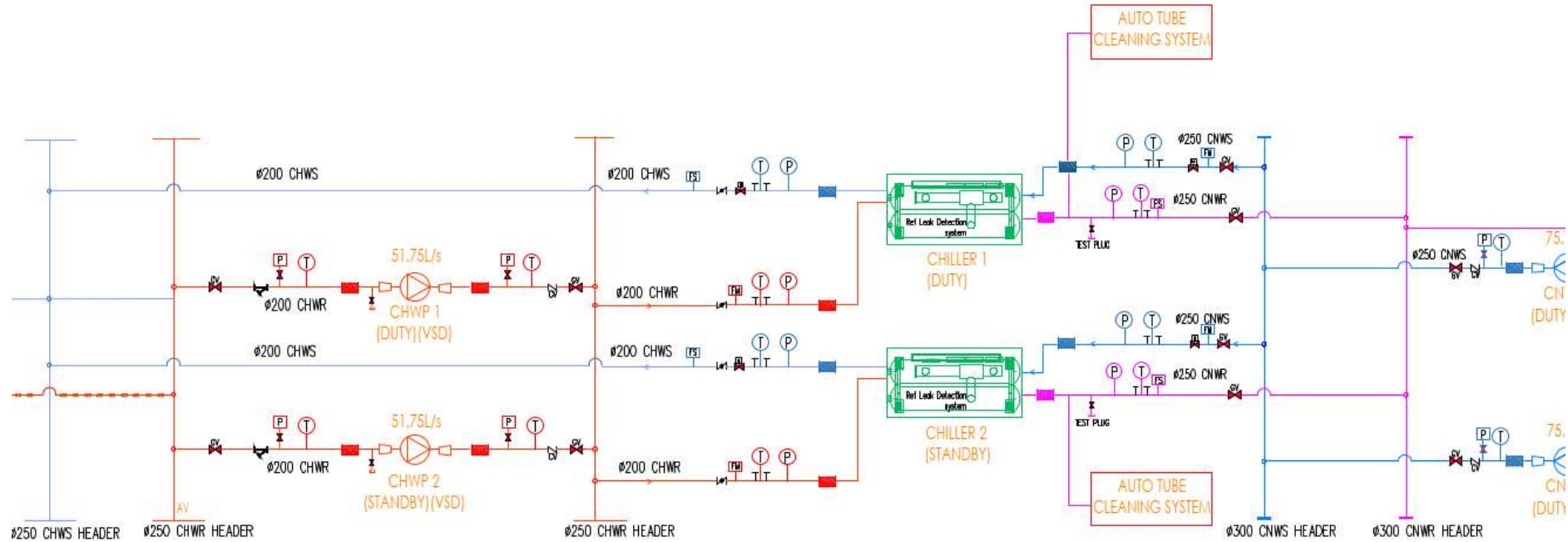
Typical Pipe Connection to Chiller

There are 4 pipe connected with Chiller

1. Chilled Water Supply Pipe (CHWS)
2. Chilled Water Return Pipe (CHWR)
3. Condenser Water Supply Pipe (CWS)
4. Condenser Water Return Pipe (CWR)



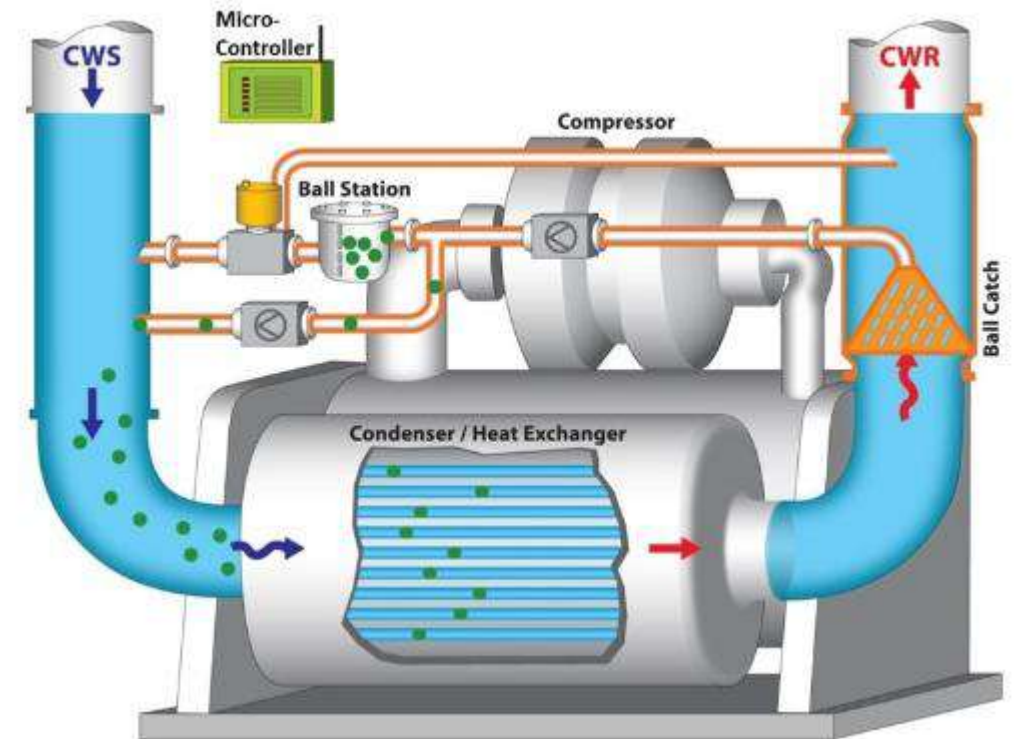
Typical Pipe Connection to Chiller



- ❖ Header Pipe : The header pipe acts as a main supply or return line that connects the chillers to the distribution system.
- ❖ Auto Tube Cleaning System is connected with condenser side of Chiller

2. Auto Tube Cleaning System

- ▶ An auto tube cleaning system for a chiller is a mechanical system that helps keep the chiller's heat transfer tubes clean and free from fouling or scaling.
- ▶ It is designed to improve the **chiller's efficiency, reduce energy consumption, and minimize maintenance requirements.**
- ▶ The auto tube cleaning system works by periodically injecting the cleaning brushes or balls into the chiller's tubes. As the chiller operates, the brushes or balls travel through the tubes, mechanically scrubbing the inner surfaces and removing any accumulated fouling or deposits.



3. Cooling Tower

Usually following are the pipe connected with Cooling Tower;

- ❖ **Condenser Water Supply Pipe (CWS)** : Supply water from Cooling Tower to Chiller
- ❖ **Condenser Water Return Pipe (CWR)** : Supply water from Chiller to Cooling Tower
- ❖ **Makeup Water Pipe**

Makeup water pipe supplies fresh water to the cooling tower system to compensate for water losses due to evaporation, drift, and bleed-off.

- ❖ **Balancing Pipe**

The primary purpose of balancing pipes is to ensure equal distribution of water flow and pressure across the cooling tower cells.

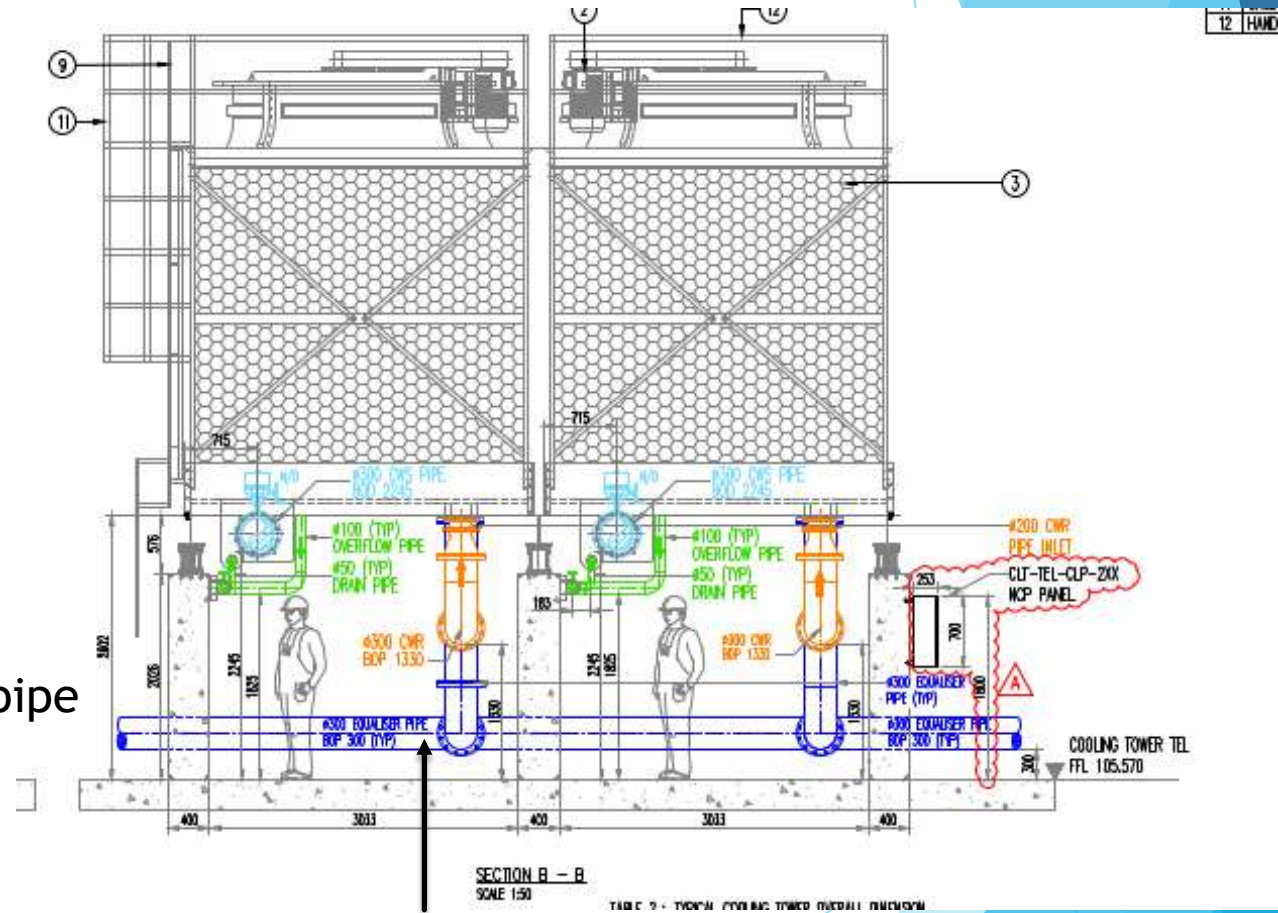
By connecting the cells together, balancing pipes allow water to be transferred between cells to balance out any differences in flow rate or pressure.

- ❖ **Overflow Pipe**

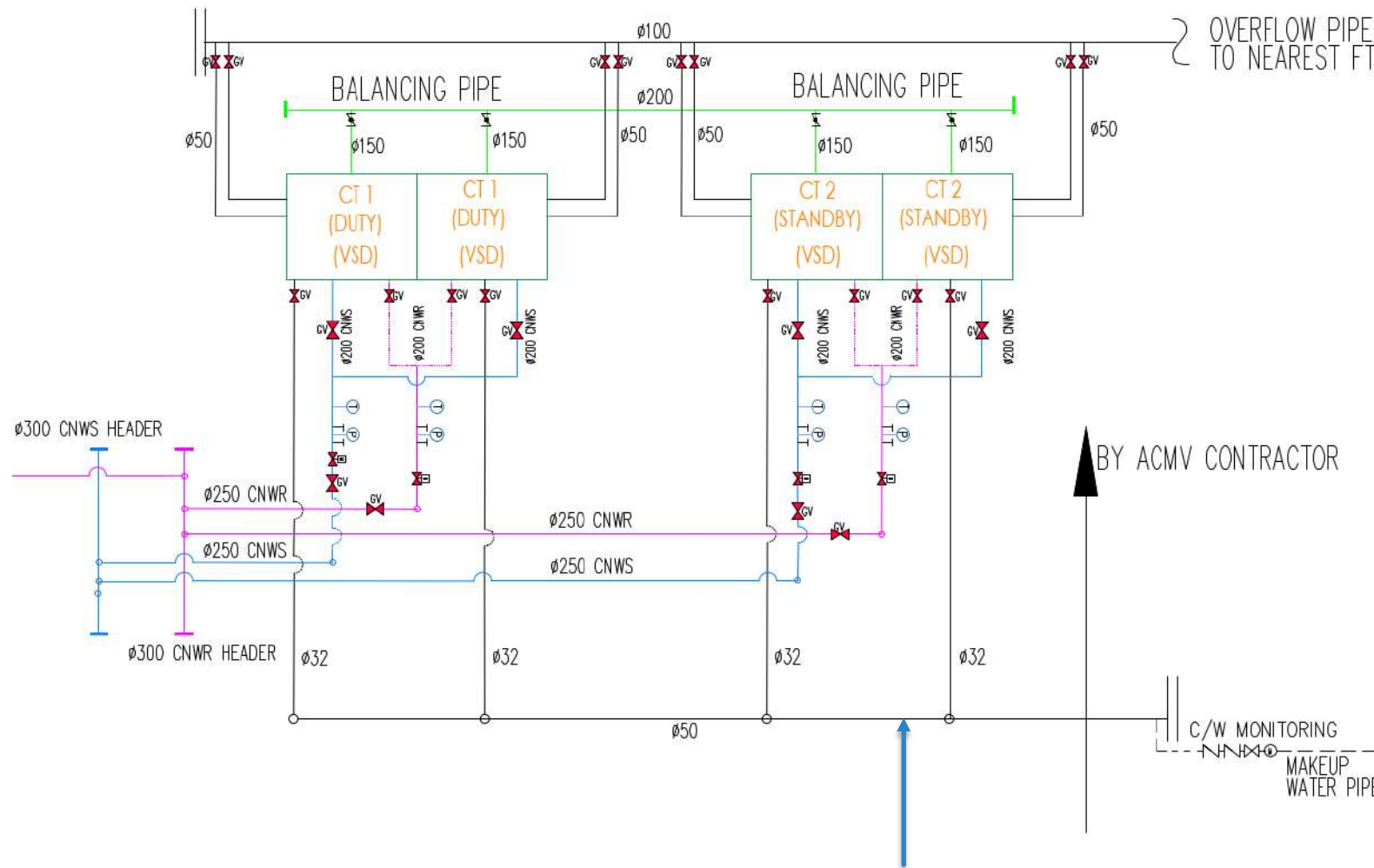
The overflow pipe in a cooling tower serves as a safety mechanism to prevent overfilling and excessive water levels within the tower.

The purpose of overflow pipe is designed to redirect excess water from the cooling tower to a safe location, such as a drain or an external water collection system. It helps prevent water from overflowing and causing damage to the cooling tower or the surrounding area.

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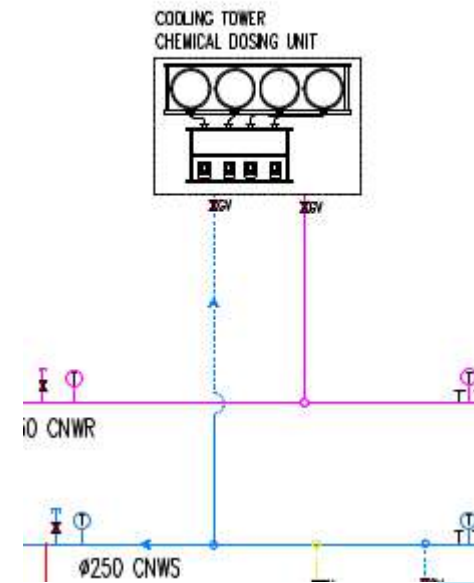
Typical Pipe Connection to Cooling Tower



Make Up Water Pipe

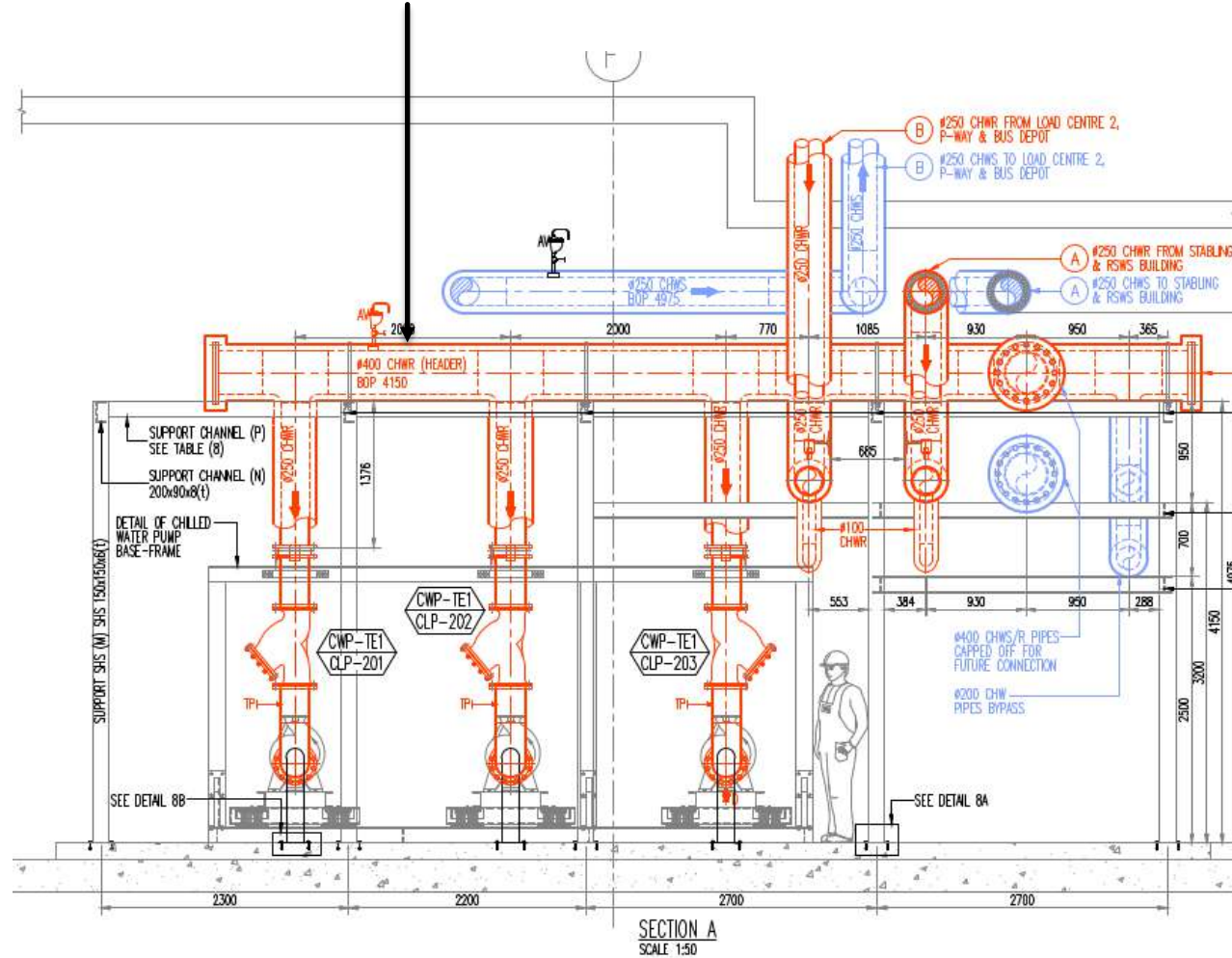
4. Chemical Water Treatment for Cooling Tower

- ▶ Chemical water treatment for cooling towers is a process that involves the use of chemicals to treat the water in a cooling tower system.
- ▶ Its is connected with CWS & CWR pipe as shown in the picture
- ▶ The primary goals of chemical water treatment are to **prevent scale formation, control corrosion, inhibit microbial growth, and maintain water quality.**
- ▶ By implementing proper chemical treatment, the cooling tower system can operate efficiently, minimize maintenance issues, and ensure the safety and comfort of building occupants.

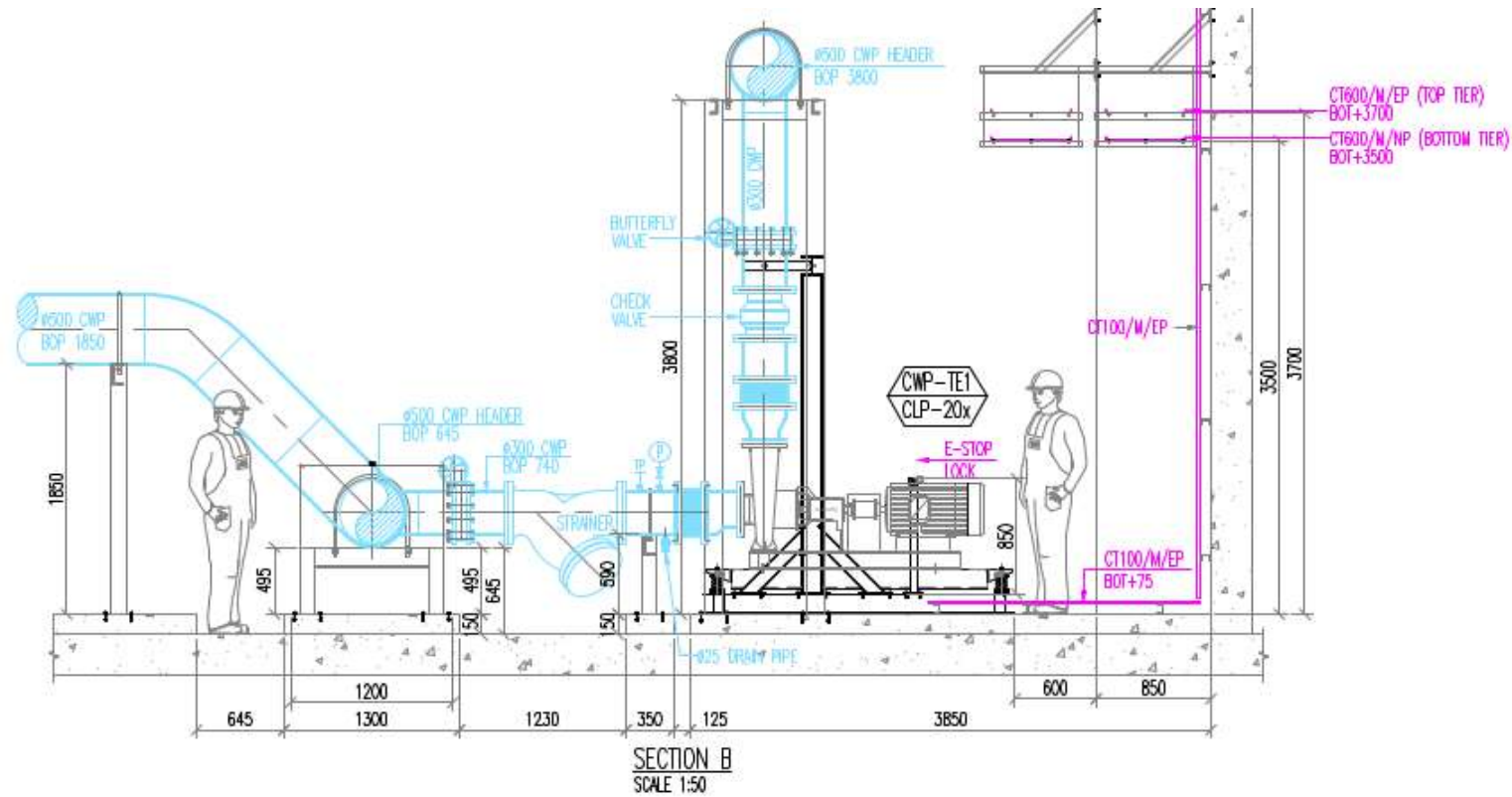


5. Typical Pipe Connection to Chilled Water Pump

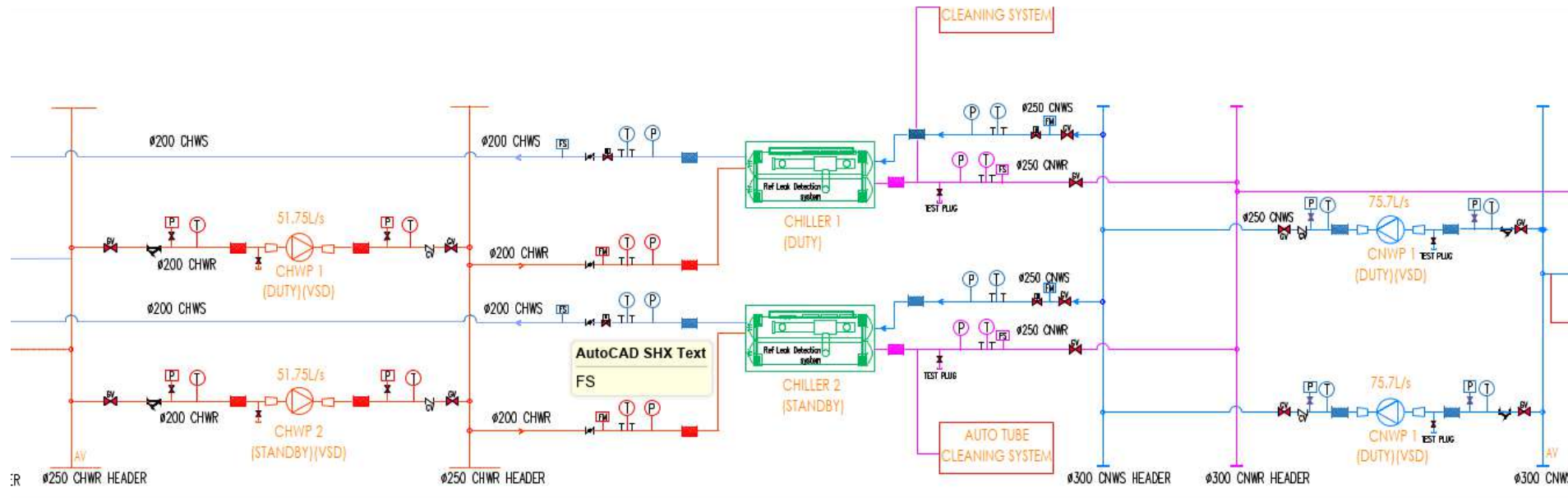
CHWR Header pipe



Typical Pipe Connection to Condenser water Pump



Typical Pipe Connection to Pump

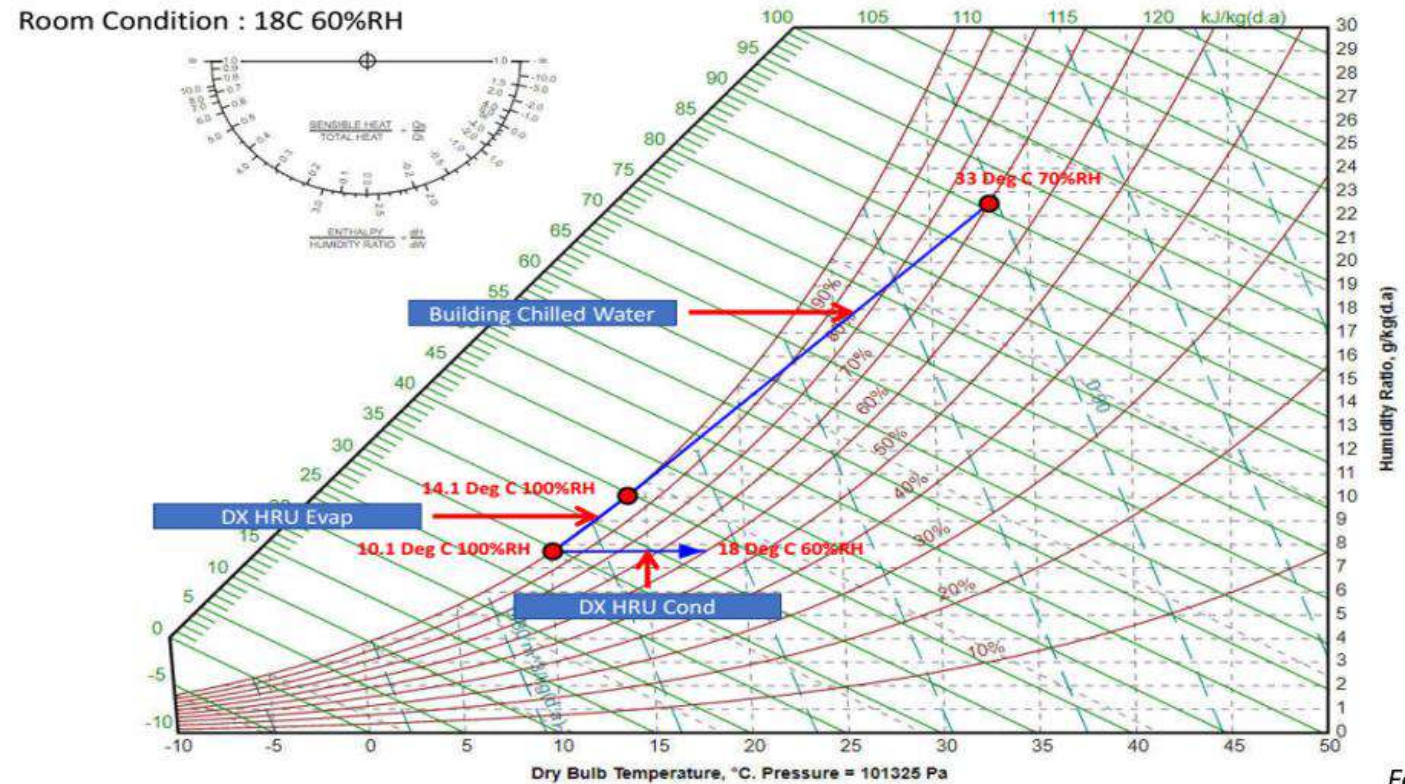


Heat Recovery Unit (HRU)

Heat recovery unit (HRU) is mostly used to achieve the room condition RH %.

There are two type of HRU

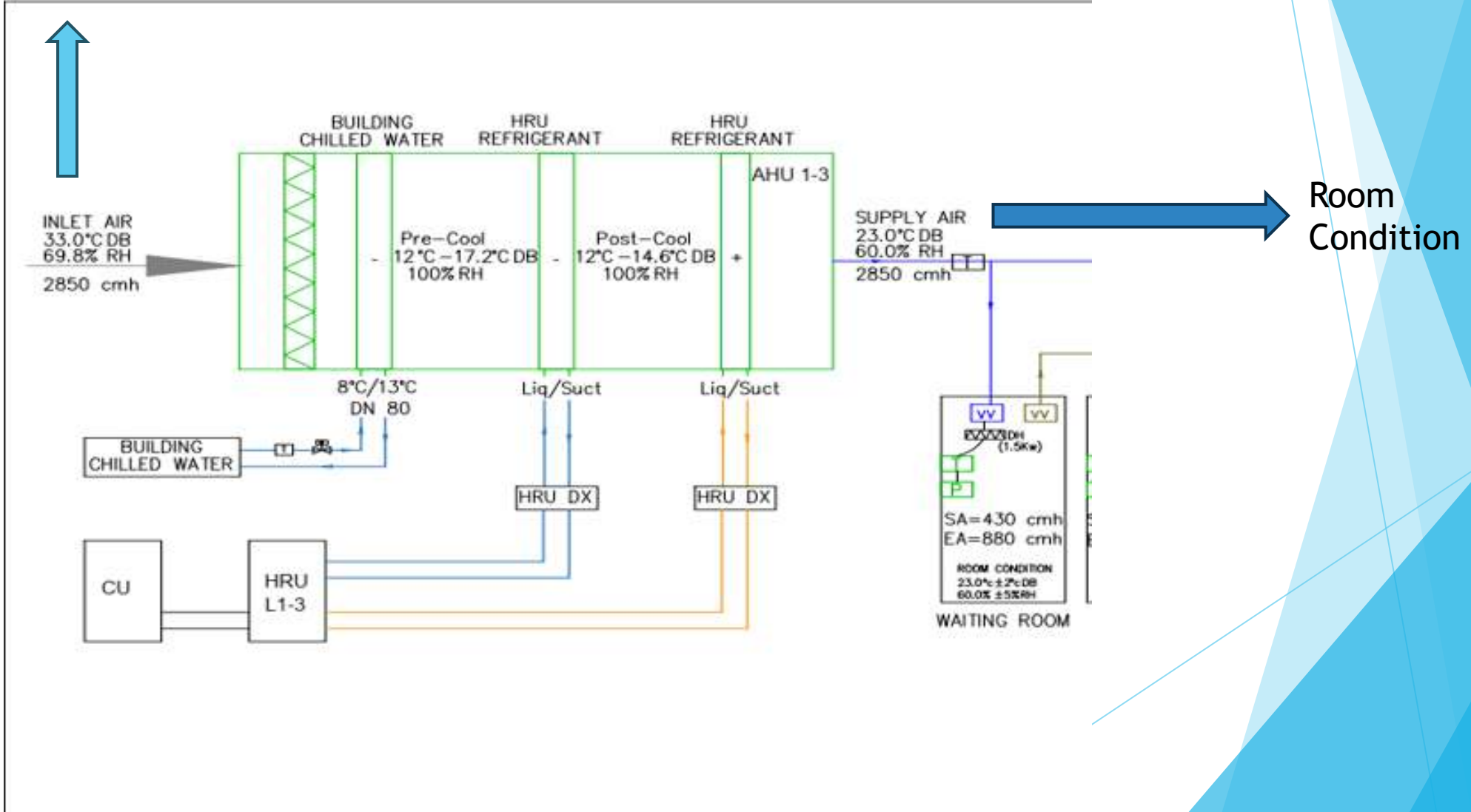
1. Chilled water HRU
2. Refrigrant HRU



For Illustration Purpose

AHU & HRU pipe connection

Out door air



Thank you !